PALEOMAGNETISM IN ABITIBI GREENSTONE BELT, AND ABITIBI AND MATACHEWAN DIABASE DIKES: EVIDENCE OF THE ARCHEAN GEOMAGNETIC FIELD[1,2]

E. IRVING AND A. J. NALDRETT

Earth Physics Branch, Department of Energy, Mines and Resources, Ottawa and Department of Geology, University of Toronto, Toronto

ABSTRACT

Paleomagnetic results are described from three Archean rock units, and two sets of diabase dikes that intrude them. The magnetization of the country rocks is reset by intrusion of the diabase up to distances of about 80 m from contacts. The temperature reached during intrusion at this distance determined from thermal demagnetization is about 400°C, which is also the approximate temperature predicted from Jaeger's cooling curves assuming intrusion into cool rock. At greater distances other and presumably older magnetizations are found with a westerly direction. The oldest dykes (the Matachewan) have an age of −2.6 Ga so that this westerly directed magnetization was acquired during the Archean. Its age, other than the fact that it is Archean, is not known, but by comparison with results from the Stillwater Complex it is estimated to be about −2.8 Ga. The results are used along with previously published data to define in tentative fashion an apparent polar wandering curve for the late Archean and early Aphebian which is referred to as Track 6. Track 6 has a simple form, and there is no indication that the horizontal mobility of the crust in the late Archean was any greater than during the Proterozoic.

INTRODUCTION

The purpose of this paper is to describe new paleomagnetic results from the Abitibi Greenstone Belt, and then to use them to construct a path of apparent polar wander (APW) for late Archean and early Aphebian time (approximately −2.3 to −2.8 Ga, Ga = 10^9y). This polar path is of potential importance for stratigraphic correlations and for the tectonic history of late Archean time. For instance, it would provide information relating to the nature of the Kenoran Orogeny. Sampling was carried out during the summers of 1970, 1971, and 1973.

The Abitibi greenstone belt is relatively weakly metamorphosed, the grade of metamorphism varying from lower greenschist to prehnite-pumpellyite facies (Jolly 1974). In Munro township, where much of our work has been done, it is prehnite-pumpellyite (Arndt et al. 1977). Because the rocks have undergone only low-grade metamorphism it does not necessarily follow that their original magnetizations have been retained. Rocks that are essentially unaltered geologically, can have had their original magnetization replaced partly or entirely by secondary magnetization. This phenomenon of partial or complete resetting of magnetizations is widespread in Precambrian terrains. (See documentation of magnetic overprints, figs. 9, 10, and 11, Irving and McGlynn 1976.) The temperatures achieved during low-grade metamorphism (say 300°C) are lower than the blocking temperature of rocks commonly observed on laboratory time scales (typically 500 to 600°C), but magnetization can be destroyed if these temperatures are applied for long intervals of time. Time itself becomes an effective agent of demagnetization. Such secondary magnetization has been shown by blocking temperatures, ranging from the Curie temperature of the constituent minerals down to room temperatures, whereas the magnetization that has survived is characterized by a small discrete range of blocking temperatures within 50° or so of the Curie temperature. The theoretical explanation of this phenomenon has recently been given by Buchan and

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Dunlop (in Pullaiah et al. 1975). The magnetization of single-domain pure magnetite and hematite should survive the prehnite-pumpellyite and low greenschist facies metamorphism provided there has been no chemical change of the iron minerals. It ought to be possible to retrieve original magnetizations in the Abitibi Greenstone Belt provided that magnetizations with high discrete blocking temperatures (that is square-shouldered thermal decay curves) can be located. The key observations required are the thermal decay characteristics.

The metamorphism associated with the Kenoran Orogeny, is however not the only possible cause of remagnetization. Long-term mild reheating may have accompanied either the intrusion of Proterozoic dikes, or burial beneath Huronian sediments which may once have blanketed this area. Thus studies of dike contacts have been made, and one of the by-products of this work has been that we have been able to show, for the first time, that the characteristic magnetization directions of the Abitibi and Matachewan dike swarms were acquired at the time they were intruded.

The first important study of palaeomagnetism of Archean rocks of Laurentia was made by Berg (1970) on the Stillwater Igneous Complex. Estimates of the age of emplacement of the complex have been obtained by Nunes and Tilton (1971) who obtained -2.75 Ga by the U-Pb method from zircons in the basal zone, and by Mueller and Wooden (1976) who obtained -2.75 ± 0.05 Ga for an Rb-Sr isochron from the contact aureole. (Rb-Sr ages in this paper are quoted for ^{87}Rb decay constant of $1.39 \times 10^{-11} y^{-1}$.) Berg obtained some consistent magnetization directions, but the layering at his sites is approximately uniform so that it is not possible to use Graham's fold test (Graham 1949) to determine whether the magnetization was acquired before or after deformation. Berg believed that the magnetization was acquired before deformation. It is equally possible that the magnetization was acquired after deformation. For example the thick Archean sills of the Chibougamau Greenstone Belt have magnetizations which structural evidence shows were acquired after deformation (Ueno and Irving 1976). Ueno and Irving therefore considered these magnetizations to be somewhat younger than the Kenoran Orogeny, which in this area is determined by Rb-Sr isochrons to be about -2.6 Ga (Jones et al. 1974). Ridler and Foster (1975) have given results in abstract from the Skead Group. Although they do not report any thermal studies, they do report that they have carried out a fold test indicating that the magnetization is Archean in age. Symons (1967) has reported work on the Ely greenstones of Minnesota but no thermal studies were made and only nine samples studied; the data do not fulfil the minimum reliability criteria of Hicken et al. (1972) and are therefore not considered further in this paper. Preliminary results have been reported from the Shebandowan volcanic-plutonic belt at the Superior Geotraverse Conferences by Dunlop and Buchan (1975, 1976) and by Halls (1975), and abstracts of work by Londry and Symons (1976) and Dunlop and Buchan (1976) have appeared but no thermal studies have been reported, and the results are not therefore considered further in this paper.

REGIONAL GEOLOGY

Goodwin and Ridler (1970) have defined the Abitibi Greenstone Belt as a crudely parallelogram-shaped structure extending from the Grenville Front in the east through Noranda and Timmins to the belt of gneisses associated with the Kapuskasing gravity high in the west. It is bounded to north and south by areas of metasedimentary and gneissic rocks, which are intruded by a variety of felsic plutons, one of which was dated by Krogh and Davis (1971) at -2.7 Ga using the U-Pb method on zircons. Much of our work has been in Munro and Dundonald townships

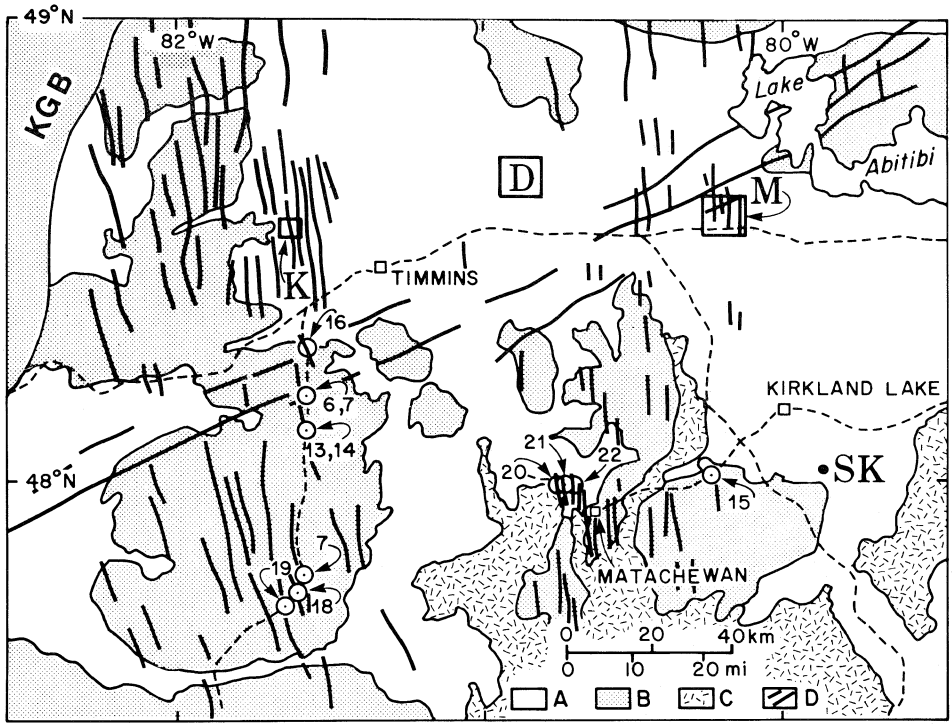


FIG. 1.—Sketch-match of sampling region taken from Ontario Geological Map No. 2198. *A* = Mafic and ultramafic intrusives, metavolcanics and metasediments; *B* = granitic rocks; *C* = Huronian sediments; *D* = diabase dikes; *KGB* = Kapuskasing gneiss belt. The northeast trending dikes are of the Abitibi swarm. The north-south dikes belong to the Matachewan swarm. The sampling sites are numbered.

in the western part of the belt (fig. 1). In the vicinity of Munro, the oldest rocks constitute a dominantly south-facing succession and occur within and to the north of the township. They belong to the Kinojevis Group (Dimroth et al. 1973; Jensen, in prep.) and consist largely of Fe-rich tholeiitic basalts with a small proportion of intermediate to felsic volcanic rock, and a succession of ultramafic lavas and mafic-ultramafic intrusions which are referred to by Arndt et al. (1977) as the Munro Formation. The Kinojevis Group is overlain by the Blake River Group, a thick succession of dominantly calc-alkaline volcanic rocks that outcrop over a large, crudely elliptical area extending from Nighthawk Lake in the west to east of Noranda. The Blake River Group is overlain by the Skead Group.

In Munro and adjacent townships the Munro Formation has been folded about east-southeast trending axes and disrupted by a series of early faults with similar strike. Gabbros have been injected along these faults and, at first sight, appear to be concordant, although Arndt (1976) has demonstrated that they cut across concordant strata of the Munro Formation at the noses of folds. These fault-controlled gabbros are cut by north-south dikes of Matachewan age (-2.69 ± 0.93 Ga, Gates and Hurley 1973) which, in turn, are cut by east-northeast trending Abitibi dikes. The latter have yielded an Rb-Sr isochron of -2.15 ± 0.68 Ga (Gates and Hurley 1973).

The Dundonald sill (Naldrett and Mason 1968), sampled by us in Dundonald township 50 km west of Munro, is identical to the sills in Munro township (MacRae 1969), and probably lies within the Munro Formation, although poor outcrop and complex structure in the intervening area leave this an open question. The Kamiskotia complex sampled in Robb township 15 km northwest of Timmins, differs from the sills of Munro and Dundonald townships in that it contains less peridotite and pyroxenite and more anorthositic gabbro. It is intrusive into a volcanic succession that is capped by felsic rocks. Poor outcrop has prevented correlation of this succession with rocks east of Timmins. The Kamiskotia complex belongs petrologically to a group of anorthositic magnetite rich intrusions including the Doré Lake and Bell River complexes (Class I subclass ii(b) of Naldrett and Cabri 1976), which are probably comparable in age to the basement.

METHODS

Samples were cored by gasoline drill and oriented by magnetic and solar compass. About five cores were drilled at each collecting site, and two cylindrical specimens were cut from each core. The remanent magnetization of the specimens was measured by spinner magnetometers. All specimens were demagnetized by step-wise heating, or by alternating fields (a.f. peak field strength are quoted) using the apparatus described by Roy et al. (1969, 1971, 1973). Demagnetization was carried in fields of less than two gammas. In some samples, particularly from the Dundonald sill, no coherent magnetization could be found; if the directions in specimens from the same sample disagreed by more than 40°, the results were rejected. Eight percent of samples collected were rejected on this basis. Results from each specimen are averaged to give means for each core. The core means are averaged giving each unit weight to obtain the mean for each sampling site. The overall averages for each rock unit are calculated giving unit weight to the mean of each sampling site.

ABITIBI DIKES

In their unaltered state these dikes consist of 50–60 modal percent plagioclase, 30% augite, 5–10% quartz and 5–7% opaques. The plagioclase is fresh and occurs as stubby to acicular laths up to 2 mm long, bearing an ophitic to subophitic relationship to the enclosing augite, which is mostly replaced by chlorite, brown and green amphibole, and biotite. Quartz occurs interstitially to the plagioclase and pyroxene. Magnetite was the original opaque mineral occurring as equant grains up to 1 mm in diameter with ilmenite exsolved around the margins and parallel to [111]. In most samples this primary magnetite has been totally altered, leaving only the ilmenite lamellae with a matrix of very fine-grained poorly reflecting material that could not be identified in polished section. Small amounts (0.5%) of pyrite are present, commonly associated with biotite. Trace quantities of pyrrhotite and chalcopyrite are also observed.

A study of the natural remanent magnetization (n.r.m.) of the Abitibi dikes has been described by Larochelle (1966). He reported a dominant westerly magnetization (*W*) with steep inclination from sampling sites immediately to the northeast of figure 1. In addition he reported two subordinate magnetizations, with directions to *NNE* and *SW*. The *W* magnetization has been assumed by most workers (e.g. Robertson and Fahrig 1971) to be the primary magnetization. McGlynn et al. (1975) and Pullaiah and Irving (1975) have recently questioned the age of magnetization of the Abitibi dikes, and Roy and Lapointe (1976) have argued that the similarity between directions of the *NNE* and *SW* magnetizations and what they regard as a secondary magnetization of the Nipissing diabase, suggests that they too are really secondary mag-

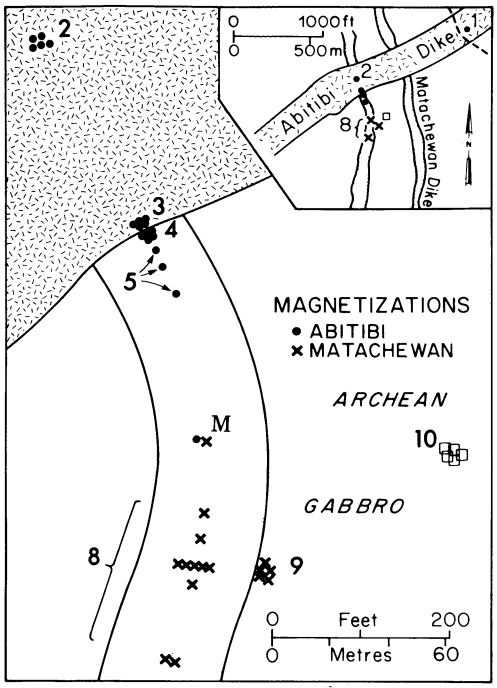


FIG. 2.—Sampling of dikes and contacts in Munro Township. The magnetizations are distinguished by different symbols. At site *M* both Matachewan and Abitibi magnetizations are found.

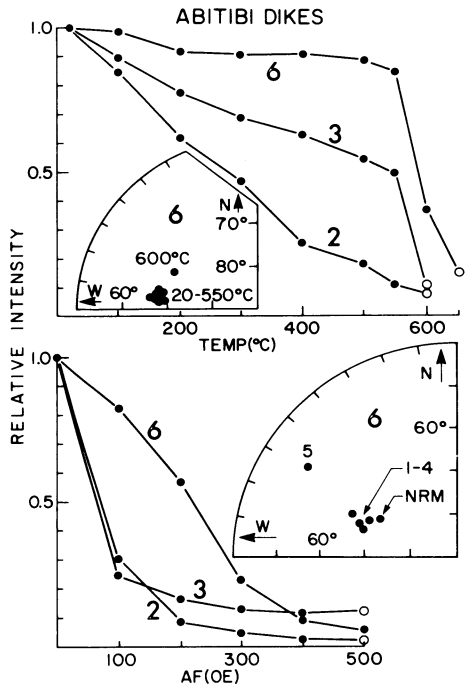


FIG. 3.—Demagnetization of Abitibi diabase. Thermal (*above*) and a.f. (*below*) demagnetization of specimens from three collecting sites. The corresponding direction changes for site 6 are shown in the insets which are parts of an equal-area net. The directions are all downward. When very large change in direction occur, the experimental points on the decay curves are marked by open symbols at which stage the n.r.m. has been destroyed.

netizations. The *NNE* and *SW* magnetizations could be a "Hudsonian" overprint acquired at about 1.8 Ga. The main problem however is to determine if the dominant *W* magnetization of the Abitibi dikes is truly the original magnetization acquired at the time the dikes were intruded. We now consider this question.

Samples were taken from two dikes and from their baked margins at sites 6 and 7 and (fig. 1) and in Munro Township (fig. 2). Both dikes have the typical *W* magnetization observed by Larochelle (1966). The thermal demagnetization curves, show wide variations, some diabase specimens (site 6) have discrete blocking temperatures, others (site 3 and 2) have blocking temperatures distributed between room temperature and 500°C (fig. 3). During a.f. demagnetization the directions of site 6 remain unchanged up to 400 Oe, but in higher fields they soon become random. During thermal demagnetization the directions at site 6 remain constant up to 550°C, are not greatly changed at 600°, but are random at 650°. The corresponding decay curve is square-shouldered, with blocking temperatures predominantly between 550 and 650°C. This indicates that magnetite is the main carrier, but that hematite is present also in small quantities. This is confirmed by studies of polished sections from site 6. Large grains of magnetite are seen to contain very fine exsolved lamellae of ilmenite, rims of ilmenite, and also fine exsolution lamellae

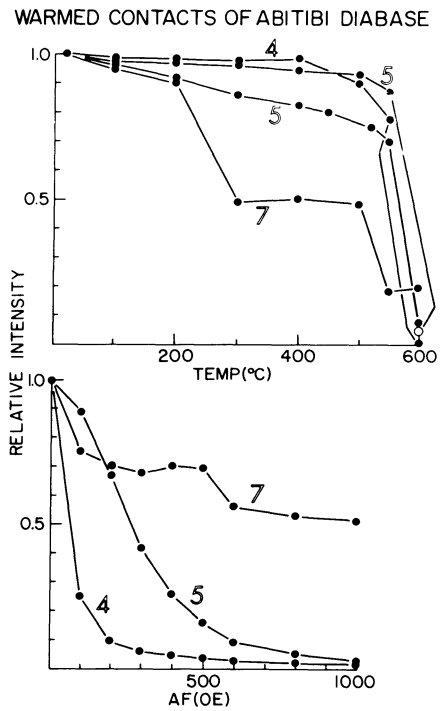


FIG. 4.—Demagnetization of rocks near contact of Abitibi diabase. The remanence has been reset and the variations of direction during demagnetization are small comparable to those shown in Fig. 3. Sites 4 and 5 are Matachewan diabase, and site 7 is the granite in contact with the diabase at site 6.

of hematite. The directions in the baked granite adjoining the diabase dike at site 6 are parallel to those in the diabase (compare means at sites 6 and 7 in table 1), and the remanent coercivities exceed 1,000 Oe (fig. 4). The blocking temperatures are distributed between 300 to 600°C with a small component retained at 600°C. This indicates that fine-grained hematite is the carrier of the remanence. In polished section the oxides are seen to consist of 0.5 to 1 mm diameter magnetite grains, now partially altered, and with ilmenite lamellae oriented parallel to (111). Chromite also occurs in association with altered sphene.

The dike in Munro Township has been studied in more detail at a point where it cuts a N-S trending Matachewan dike which is itself cutting one of the gabbroic bodies referred to above as having been emplaced along an ESE-striking fault (fig. 2). Samples have been taken from the dike itself (sites 1, 2, and 3) from its contact with Matachewan diabase and outwards along the Matachewan dike for a distance of 150 m. The remanent coercive forces in this Abitibi dike are rather low, the median destructive field (m.d.f.) being less than 100 Oe (fig. 3, sites 2 and 3). The blocking temperatures are generally distributed between room temperature and 600°C, with a concentration between 550°C and 600°C in specimens at the chilled contact (fig. 3, site 3). This indicates that magnetite or titano-magnetite are carriers of the remanence. In polished section, much of the magnetite has been altered to a fine-grained poorly reflecting material containing lamellae of original exsolution ilmenite, but localized areas of the original magnetite remain unaffected. In addition ultrafine (1 micron) blebs of an oxide that may be magnetite, but which are too fine to identify with certainty, occur along the cleavage planes of altered pyroxene. Warmed Matachewan diabase within 30 m of the contact has the same *W* direction, and square-shouldered thermal decay curves with blocking temperatures between 550 and 600°C (fig. 4, sites 4 and 5). High remanent coercive forces are present and the directions are little changed in fields up to 1,000 Oe. At site *M* (fig. 2) 80 m from the contact two specimen cores were collected within 5 m of one another. One gave a direction 290, +63 and the other a direction 209, +03 corresponding to the Abitibi *W* and Matachewan (see below) directions respectively. Traced southward along the Matachewan dike the *W* magnetization is retained for a distance of 80 m, beyond which typical Matachewan magnetizations occur. Clearly during the intrusion of Abitibi diabase, remagnetization on a regional scale did not occur, but was confined locally to a zone about 80 m wide

adjacent to the diabase itself. It follows also that the *W* magnetization was acquired during cooling, and is the primary magnetization as previous workers have assumed it to be. Using the above thermal demagnetization results and the approximations of Pullaiah et al. (1975), the temperature attained at a distance of about 80 m may be estimated to be about 400°C. Consideration of Jaeger's (1957) cooling curves applied to a diabase dyke equal in thickness to the dike studied by us (100 m) and intruding material of similar conductivity and density to itself, suggests that the maximum temperature reached at a point 80 m from the contact would be 360°C higher than the temperature of the country rocks at the time of intrusion. Provided that the temperature of the country rocks was low, this is consistent with the conclusions from the magnetization studies. The change from Abitibi to Matachewan directions apparently occurs suddenly within a matter of metres since at site *M* of figure 2 both magnetizations occur. There is no evidence of a transition zone where both magnetizations occur, and in neither sample is there a change during demagnetization from *W* Abitibi to Matachewan directions; the magnetization appears to be either completely Abitibi *W* or completely Matachewan. This is consistent with the observed presence of very narrow discrete range of blocking temperatures. Unfortunately, because of heavy bush, sampling on either side of site *M* was not practicable.

Mean directions in all cores from Abitibi diabase and contact rocks are shown in figure 5, and summarized in table 1. The directions in contacts and diabase are in excellent agreement.

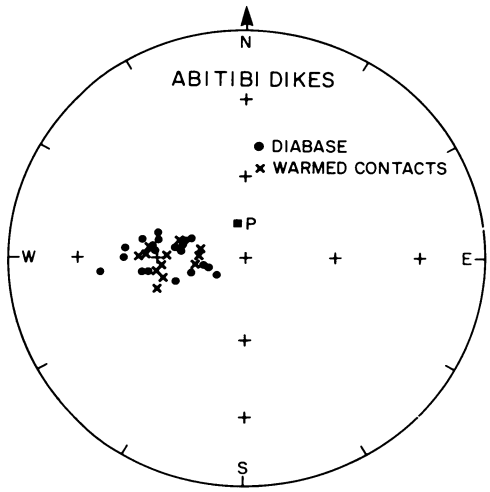


FIG. 5.—Directions observed in core samples (mean of two specimens from each core) from Abitibi dikes and warmed contacts rocks. The primitive is the present horizontal. *P* is the direction of the present field. All directions are downwards.

MATACHEWAN DIKES

The Matachewan diabase dikes form an extensive swarm of north-trending vertical dikes (fig. 1). The samples studied by us resemble the Abitibi dikes in that they are composed of 50–60% plagioclase, 30% augite, 5–10% quartz and granophyric intergrowths, and 5% opaques. The plagioclase is lath-like, but the augite differs from that in the Abitibi dikes in that in addition to developing interstitially to the plagioclase, much of it occurs also as equant grains. The principal oxide mineral is magnetite, now largely altered to a fine grained poorly reflecting material, and containing ilmenite lamellae and rims.

The remanent magnetization of the Matachewan dikes has been studied previously by Strangway (1964) and Fahrig et al. (1965) who carried out magnetic cleaning in fields of about 100 Oe, but who reported no thermal or contact studies. These workers found directions of opposite polarities, but they were not antiparallel, the misalignment being on average 40°, and in the sense expected from incomplete removal of viscous magnetizations caused by the present geomagnetic field. Incomplete cleaning in these early studies is probably the cause of the misalignment, because we find that after cleaning in fields of 300 or 400 Oe there is no evidence of misalignment.

TABLE 1

RESULTS FROM EACH SAMPLING SITE

Site n		M_n	D, I	R	$\delta°$	$\alpha_{95}°$	Treatment
Abitibi dikes (d) and contacts (c):							
1(d)	4	3111	272, +51	3.90	13	17	300–800 Oe
2(d)	5	1766	255, +71	4.94	9	9	200–300 Oe
3(d)	5	1853	273, +57	4.98	6	6	300 Oe
1+2+3	14	...	268, +60	13.59	14	7	200–800 Oe
4(c)	5	3348	272, +58	4.99	4	4	300 Oe
5(c)	3	7639	254, +60	3.00	3	5	300 Oe, 550°C
4 5	8	...	266, +59	7.96	6	4	300 Oe, 550°C
1 to 5	22	...	268, +60	13.59	14	7	200–800 Oe, 550°C
6(d)	6	2000	285, +66	5.99	4	4	300 Oe
7(c)	4	325	278, +72	3.99	4	5	300 Oe
6 and 7	10	...	283, +69	9.96	5	3	300 Oe
Matachewan dikes (d) and contacts (c):							
8(d)	10	3947	201, −11	9.81	11	7	300 Oe
9(c)	5	2864	208, −17	4.98	6	6	300 Oe
8 and 9	15	...	203, −13	14.72	11	5	300 Oe
11(d)	5	6739	029, +18	4.97	6	6	300 Oe
12(c)	6	1165	021, +26	5.71	18	17	300 Oe
11 and 12	11	...	025, +22	10.61	15	9	300 Oe
13(d)	7	6484	206, −29	6.98	5	5	300 Oe
14(c)	1	102	220, −14	...	...	...	300 Oe, 500°C
15(d)	5	285	210, −18	4.86	13	14	300 Oe
16(d)	5	57	220, −29	4.94	9	10	300 and 400 Oe
18(d)	5	83	215, −19	4.98	5	6	300 Oe
19(d)	3	45	200, −02	2.98	6	11	300 and 400 Oe
20(d)	5	143	210, −16	4.95	8	8	300 Oe
21(d)	5	39	203, −08	4.45	27	31	300 Oe
22(d)	5	149	201, −06	4.98	6	6	300 Oe
Archean gabbro (10) and lavas (23):							
10	4	...	189, −23	3.41	16	17	300 Oe
23	22	2941	032, −87	21.1	16	6	300 and 400°C, 500°C
Kamiskotia complex:							
24	4	2810	289, +18	3.98	5	7	300 Oe
25	4	8159	292, +23	3.92	11	15	300 Oe
26	4	1730	292, +24	3.98	4	5	300 Oe
27	1	4186	274, +26	...	...	...	300 Oe
28	3	3830	290, +21	2.98	6	11	300 Oe
29	4	1251	294, +22	3.98	7	9	300 and 400 Oe
30	6	4138	293, +23	5.98	9	8	300 Oe
Dundonald sill:							
31	5	2922	283, +41	4.81	16	17	300 Oe
32	5	5	284, +31	4.84	15	16	300 Oe
33	3	17	286, +30	2.99	5	10	300 Oe
34	1	908	245, +24	...	...	...	300 Oe
35	2	11	250, +28	1.96	...	...	300 Oe

NOTE.—n is the number of oriented cores, D, I are the declination and inclination of the mean direction, M_n the mean intensity of n.r.m. in microganss, R is the resultant, $\delta°$ the standard deviation, and α_{95} the circle of confidence (P = 0.05). The cleaning treatment used is given in the right-hand column.

We have sampled Matachewan diabase at 11 sites (figs. 1, 2, and 9). Ten sites have southerly directions and one site (site 11) has northerly directions. Typical results from demagnetization of rocks with southerly magnetizations are shown in figure 6. At site 8 in Munro township (fig. 2), the diabase samples have square-shouldered demagnetization curves with blocking temperatures between 500 and

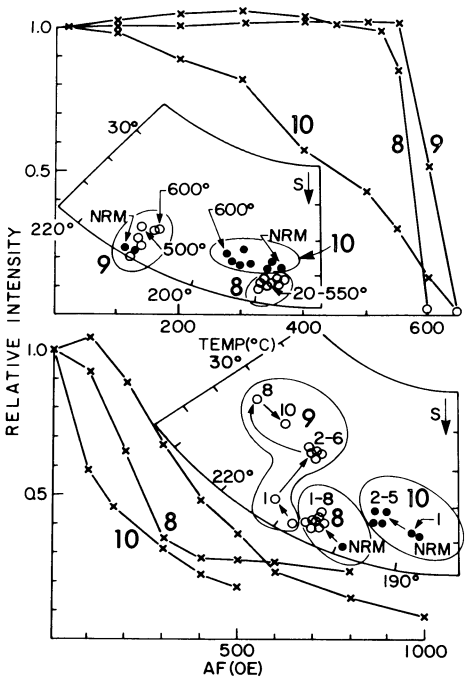


FIG. 6.—Demagnetization of Matachewan diabase (site 8), baked Archean gabbro adjacent to diabase (site 9), and Archean gabbro 65 m from diabase (site 10). See Fig. 2. Thermal decay and corresponding directions are given above, and a.f. decay and corresponding directions below. In the insets dots (open circles) denote downward (upward) inclination.

600°C. The a.f. demagnetization curves fall smoothly. During both a.f. and thermal demagnetizations there were small shifts in inclination, owing to the removal of magnetizations along the present geomagnetic field. The blocking temperature indicates that the magnetization is carried by fine-grained titanium-free magnetite. In polished section the oxides are seen to consist of fine-grained poorly reflective material pseudomorphing magnetite with ilmenite exsolution lamellae, very fine lamellae of magnetite localized along the cleavage planes of altered pyroxene, and a fine powder, probably magnetite, that is visible in transmitted light but not apparently in polished section.

The directions in the baked gabbro (site 9 in Archean gabbro) have an excellent end-point between 200 and 600 Oe, and the directions do not change greatly in a.f. up to 1,000 Oe (fig. 6). The baked gabbro has square-shouldered thermal demagnetization curves with some blocking temperatures exceeding 600°C. The thermal demagnetization curves of unbaked Archean gabbro (site 10, fig. 2) at a distance of 60 m from the Matachewan diabase, fall almost linearly with temperature. Opaque oxides include large skeletal grains of altered magnetite, containing patches and lamellae of ilmenite, and rare inclusions of hematite and chalcopyrite. A fine oxide, that could not be identified, is also associated with altered pyroxene. The directions however remain approximately constant and are similar to those observed in the Matachewan diabase and their immediate contacts (fig. 6, upper part).

Elsewhere there is a strong contrast between the directions of magnetization in Archean basement, and the Matachewan diabase and their warmed contacts. At site 11 the diabase intrudes the Kamiskotia gabbro (fig. 7 and 9, below). The gabbro samples were collected at distances 1 to 12 m west of the dyke contact. The directions in both the diabase and baked gabbro are slightly east of north, and are reversed (within the errors) from the southerly directions of figure 6 (table 1). The blocking temperatures are confined to a discrete interval between 550 and 600°C. During a.f. and thermal demagnetization the directions show excellent end-points. The directions in baked gabbro are close to those from the diabase (compare sites 11 and 12 in table 1). The directions in the Kamiskotia complex (see next section) are about 90° away. This again shows that remagnetization caused by intrusion of Matachewan diabase is local, and that the magnetization of the Matachewan diabase and their immediate contact rocks was acquired at the time of cooling at −2.6 Ga. The contrasting behaviour of similar directions in diabase and host rocks at Munro, and the very different directions

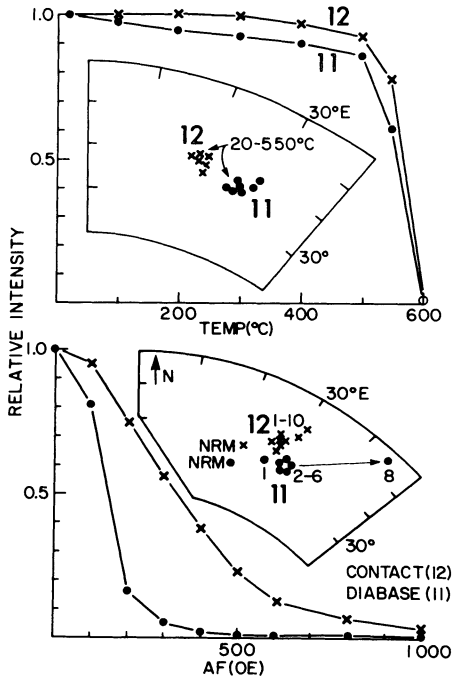


FIG. 7.—Demagnetization of normal polarity Matachewan diabase (site 11) and baked Kamiskotia gabbro (site 12). The thermal (*above*) and a.f. (*below*) demagnetization curves with the corresponding variations in direction are shown. The directions are all downwards.

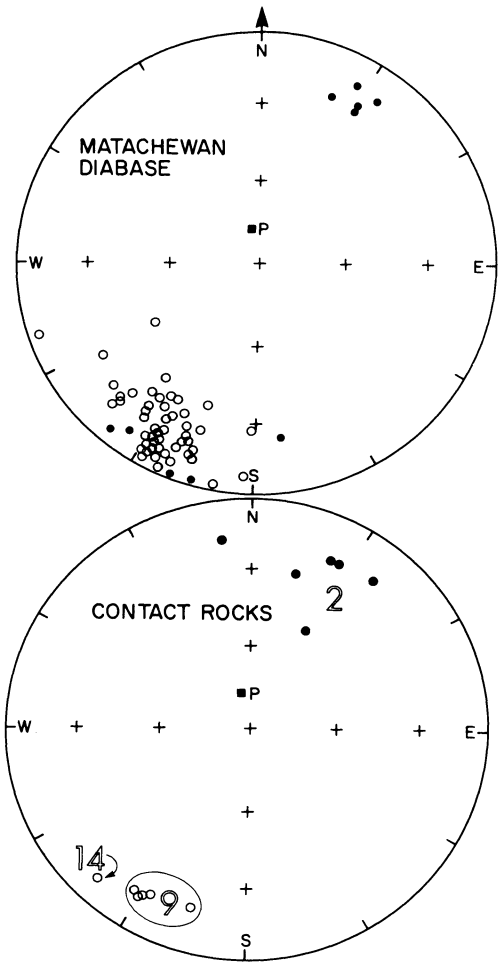


FIG. 8.—Directions observed in core samples (mean of 2 specimens from each core) of Matachewan diabase (*above*) and contact rocks (*below*) after magnetic cleaning in a.f. of 300 or 400 Oe. The warmed contact rocks are Archean gabbro site 9, Kamiskotia gabbro at site 12, and red granite at site 14. Open (solid) symbols indicate upward (downward) inclinations. The primitive in the horizontal. *P* is the direction of the present field.

at Kamiskotia, may be owing to the fact that the host gabbro at Munro, which is younger than the volcanic basement, is perhaps much closer in age to the Matachewan diabase than to the Kamiskotia gabbro.

The magnetization directions in all Matachewan diabase samples (means of two specimens) after magnetic cleaning in alternating fields of 300 or 400 Oe are shown in figure 8. The reversal is exact, within the errors, and the contact rocks are in good agreement with the diabase (table 2).

KAMISKOTIA COMPLEX

In the area sampled by us the complex is gabbroic, composed of about 50 modal percent plagioclase, 40% augite, 5% of an alteration thought to be after hypersthene, and 5% opaque oxides. The plagioclase is variable in grain size. Most grains are slightly elongated (5 mm × 1.5 m), but there are large (6 mm) equant fresh grains. Pyroxene is interstitial, enclosing plagioclase poikiolitically in some cases. The augite contains lamellae along (100)

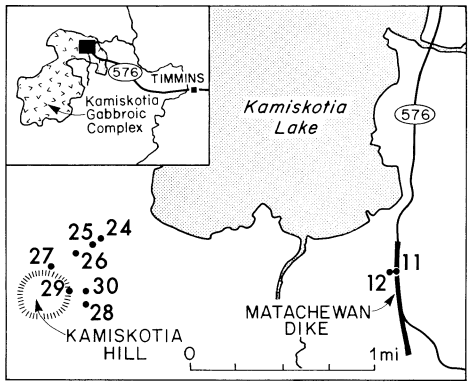


FIG. 9.—Sketch map of Kamiskotia complex showing sampling localities and dike.

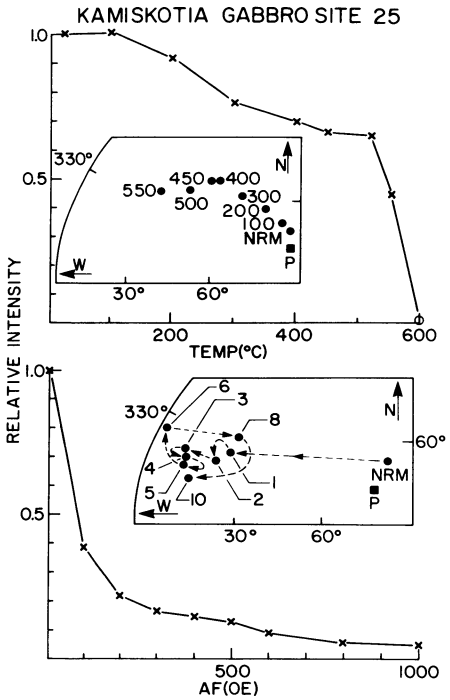


FIG. 10.—Demagnetization of Kamiskotia gabbro. Thermal (*above*) and a.f. (*below*) demagnetization curves and the corresponding changes in direction are given. Directions all downward. *P* is the present field.

and (001) of what is thought to be altered hypersthene. The oxides consist of a fine grained alteration after magnetite, with ilmenite at grain boundaries. Irregular blebs of pyrite occur along the cleavage planes of augite, together with very fine flakes of an oxide, thought to be hematite.

The intrusion is heavily timbered and not readily accessible, and water for drilling is absent over much of the outcrop. Sampling was confined to the northwestern part of the intrusion where samples from seven sites were obtained (fig. 9). Typical decay curves are shown in figure 10. The initial directions were usually near the present field, but during demagnetization they migrate to the west-northwest, and this direction is retained after demagnetization in fields up to 1,000 Oe. The thermal decay curves have a two-stepped form. Magnetizations with blocking temperatures distributed between 200 and 450°C have directions along the present field, and are presumably viscous components imposed by the present field. There is also a magnetization with blocking temperatures between 520 and 600°C and it is this that has the west-northwest directions. The carrier of the latter magnetization is evidently magnetite. After magnetic cleaning the directions in all samples are well-grouped (fig. 11), and this magnetization predates the intrusion of the Matachewan dikes because it is only immediately adjacent to the dikes that the magnetization is parallel to that in the dikes.

DUNDONALD SILL

The Dundonald sill is a simple differentiated sill ranging from peridotite at the base through pyroxenite to a gabbro which contains occasional zones of granophyric gabbro (Naldrett and Mason 1968). Since layering within the sill correlates closely with interflow contacts in the enclosing volcanics, it is thought to have crystallized with a sub-horizontal orientation, but it has since been deformed into a W-shape consisting of two synclines flanking a central anticline (fig. 12). The peridotites are mostly completely serpentinized, but pseudomorphic textures

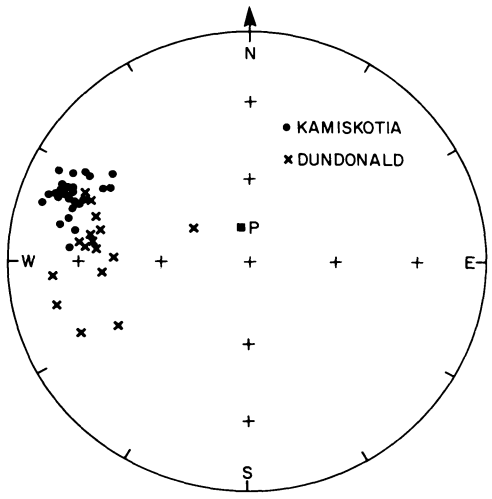


FIG. 11.—Directions of magnetization in samples (mean of 2 specimens from each core sample) from Kamiskotia complex and Dundonald sill. *P* is direction of the present field. The primitive is the present horizontal plane.

indicate that it was originally composed of olivine and augite in the proportions of 4 to 1. The pyroxenites consist of 60–95 modal percent of relatively fresh diopsidic augite with chloritic alteration after plagioclase. Fresh to moderately altered intercumulus plagioclase is common in the upper part of the pyroxenite horizon towards the contact with the gabbro, although it is usually saussuritized. The gabbros consist of cumulus plagioclase (now partially saussuritized), cumulus augite, and up to 10% of altered cumulus magnetite containing both large and extremely fine exsolution lamellae of ilmenite. The proportion of magnetite decreases upwards to less than 5% in the upper half of the gabbro.

Samples were obtained from eight sites from the north limb and three sites from the south limb (fig. 12). Opaque minerals in the peridotite are primary chromite and secondary magnetite, distributed throughout the serpentine as a fine dusting, and also as small granules concentrated along cross-cutting serpentine veinlets. Chromite is the principal oxide at the base of the pyroxenite horizon, but this gives way to magnetite in the central and upper portions of the unit. While the chromite is fresh in all samples examined, the magnetite is largely altered and contains fine lamellae of ilmenite. Fine flakes (<3 microns) of oxide, probably magnetite, occur along the cleavage in the pyroxenes. The main oxide in the gabbro is magnetite, occurring as 1 mm diameter grains containing up to 40% exsolved ilmenite, some of it exsolved on such a fine scale as to subdivide the host magnetite into areas a few tens of microns in dimensions. The magnetite is now largely altered.

Coherent magnetization was obtained from four of the five pyroxenite sites sampled and one oxide-rich gabbro site distinguished by solid dots in figure 12. During a.f. demagnetization the directions migrate away from the present field to a west-northwest direction and remain approximately constant up to 1,000 Oe (fig. 13). The thermal decay curves show a narrow range of blocking temperatures between 500 and 600°C, suggesting that fine-grained titanium-free magnetite is the carrier of the remanence. There is only one sample with coherent magnetization from the south limb, so that a fold test cannot be made. The directions in all samples after cleaning are shown in figure 11. The directions differ strongly from directions in Matachewan diabase, but are similar (although more scattered) to those of the Kamiskotia complex, indicating that the remanence is pre-Matachewan in age.

ULTRAMAFIC LAVA

Ultramafic lavas were sampled from the locality in Munro Township described by Pyke et al. (1973). Many of the flows consist of a basal zone (*B* zone) rich in cumulus olivine, overlain by a zone (*A* zone) composed of skeletal olivine and pyroxene displaying spinifex texture. Other flows do not show this zonation and are characterized by a uniform distribution of olivine phenocrysts. Although fresh olivine and pyroxene occur, secondary

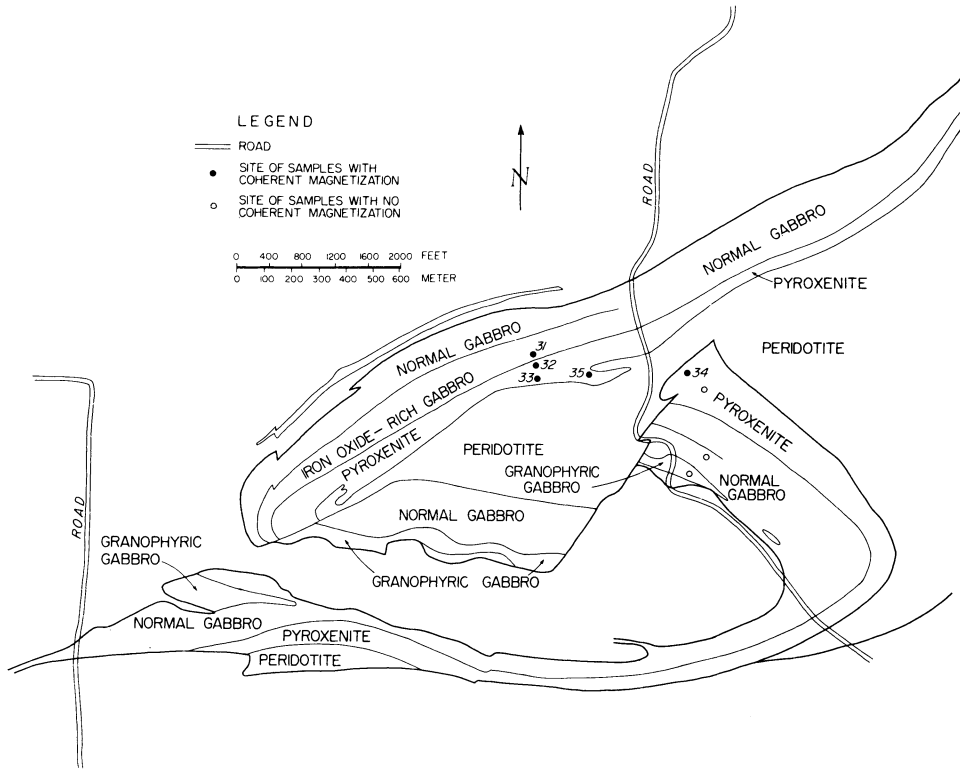


FIG. 12.—Map of the western portion of the Dundonald sill showing sampling sites. Open circles indicate sites from which no coherent data was obtained. Modified after Naldrett and Mason (1968).

alteration products, including serpentine and associated fine magnetite and hematite granules, account for over 50% of most rocks. In general, *B* zone samples tend to be highly altered, and spinifex (*A* zone) samples much less so. The lava sequence strikes east-west and dips 80°N. Sampling covered a stratigraphic interval of 70 m and included both *A* and *B* zones of the spinifex-capped flows and unzoned flows.

Magnetizations with both discrete and distributed blocking temperatures (referred to as groups I and II respectively) occur. Group I samples have blocking temperatures between 450 and 550°C, and, in specimen 26, between 550 and 650°C (fig. 14). The remanent coercive force of group I samples is rather high, the median destructive fields ranging from 200 to 400 Oe. Directional changes during magnetic cleaning are shown in figure 15 (right-hand side). In specimen 26 the directions with respect to present horizontal are initially nearly vertical and directed upward, and remain there during demagnetization up to 500 Oe. In specimen 15 the directions move into this direction after demagnetization in fields of about 200 Oe. These properties suggest that the carrier of the remanent magnetization in the group I specimens is low-titanium magnetite, or hematite in specimen 26. As mentioned above, polished section study revealed the presence of both magnetite and hematite in most samples. We were unable to distinguish any anomalous amounts of hematite in sample 26 to

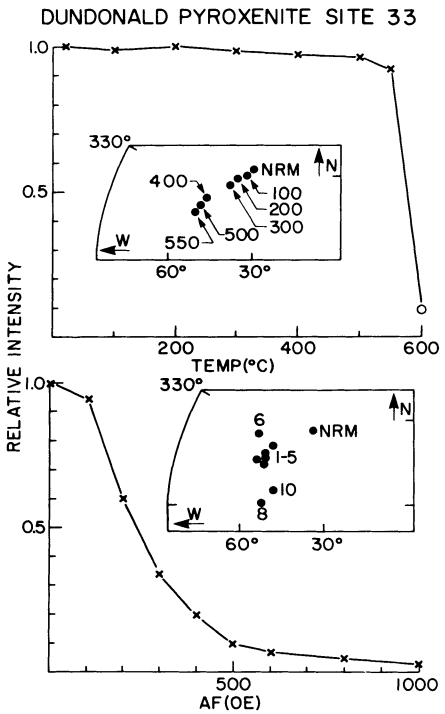


FIG. 13.—Demagnetization of pyroxenite (site 33) from Dundonald sill. Thermal (*above*) and a.f. (*below*) demagnetization and corresponding changes in direction are given.

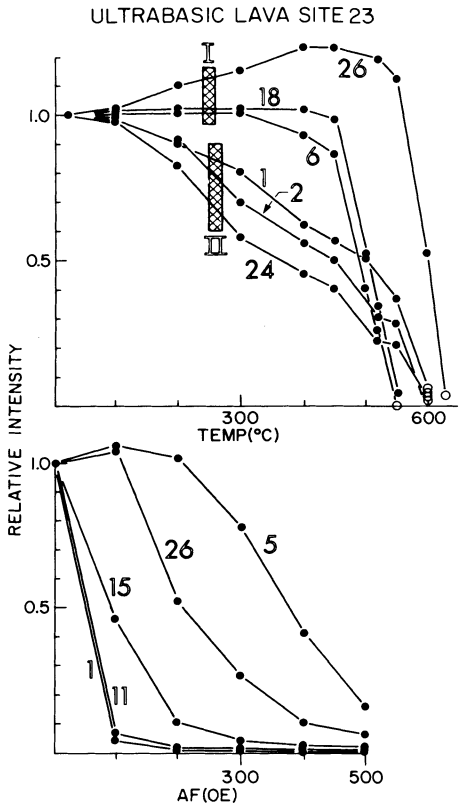


FIG. 14.—Thermal (*above*) and a.f. (*below*) demagnetization curves for ultrabasic lava, site 23. Experimental points denoted by circles indicate that very large changes in direction have occurred.

account for its markedly higher blocking temperatures. On the left-hand side of figure 15 the directions in all the group I specimens after magnetic cleaning are shown. There is a significant grouping of directions with steep upward inclination with respect to present horizontal.

Group II samples have blocking temperatures distributed between 100 and 600°C (fig. 14). Remanent coercive forces are low, and median destructive fields are about 50 Oe. The directions are scattered and have no satisfactory end-points. Vector analysis revealed no coherence among the magnetizations removed during demagnetization.

The group II magnetization are almost certainly post-Archean. The group I magnetization could by primary, perhaps acquired during serpentinization soon after deposition. There is no evidence contrary to this view. The directions agree after correction for dip with the Archean gabbro of Munro Township.

DISCUSSION

The results show that the magnetizations of the Abitibi and Matachewan dikes were acquired during cooling and have not been altered subsequently. The magnetization of the Kamiskotia complex was acquired prior to the intrusion of the Matachewan dikes and is therefore Archean. The magnetization of Dundonald sill is less easily interpreted, but the directions are similar to those in the Kamiskotia complex and therefore it is probably that its magnetization is also

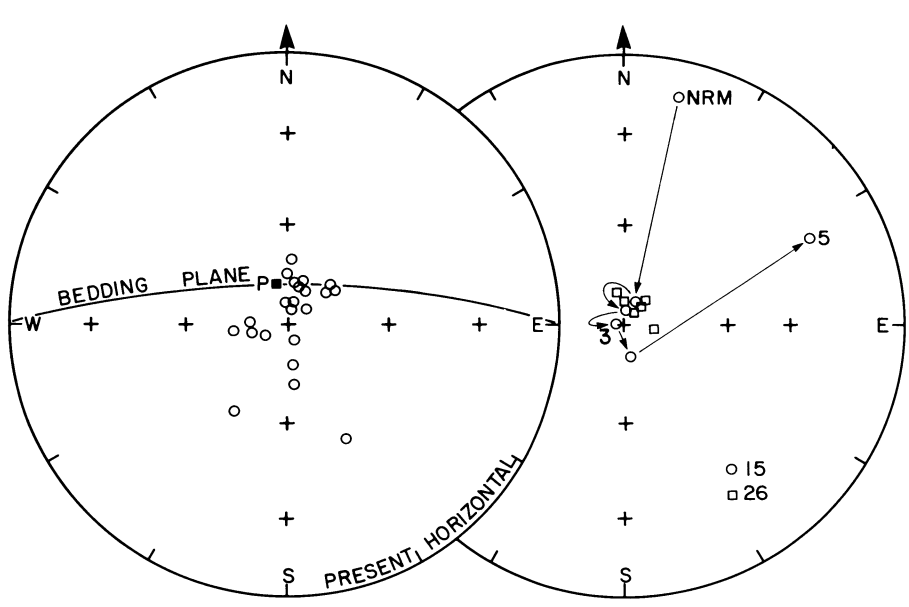


FIG. 15.—Directions of magnetization in Group I samples of ultrabasic lava, site 23. Cleaned directions in all specimens are given on the left, and typical changes in direction during a.f. demagnetization in samples 15 and 26. *P* is the present filed direction. The primitive is the horizontal.

Archean. The Dundonald sill magnetization is probably post-deformational, but this deformation may have occurred soon after formation so the magnetization may also date from that time. Rocks with high blocking temperatures in a range 500 to 600°C and square-shouldered demagnetization curves occur in the Dundonald and Kamiskotia bodies indicating that this material is unlikely to have been reset by heating during regional metamorphism to the prehnite-pumpellyite grade prevalent in this region (Pullaiah et al. 1975). The group I samples from the ultrabasic lavas give consistent directions with high blocking temperatures, but until results are available from other localities no weight can be placed on them and they are not considered further.

A summary of the results is given in table 2. The Archean poles together with those from Aphebian rocks are plotted in figure 16. A smooth curve can be drawn through them, the one exception being the pole for the Dogrib dikes which falls near Japan. A prominent bend in the curve occurs in NE Siberia at about −2.3 Ga. To conform with the classification of Irving and McGlynn (1975) we may refer to this bend as hairpin 50. The polar path following this bend is referred to as track 5 and the path before as track 6. Track 5 has been discussed previously (Irving and Lapointe 1975; McGlynn et al. 1975; Roy et al. 1975; Irving and McGlynn 1976; Roy and Lapointe 1976). Track 6 is the first recorded attempt to provide an APW curve for the late Archean based on magnetizations that are of Archean age. It is obviously very tentative indeed, both in form and timing. The age calibration at −2.6 Ga is given by the age of the Matachewan dikes. The calibration at approximately −2.8 Ga for the Dundonald sill and Kamiskotia complex is suggested, very tentatively, on the basis of the agreement of their poles with that for the Stillwater

TABLE 2

SUMMARY OF RESULTS

	B		k	$\delta_2°$	$\alpha_{95}°$
1 Abitibi dikes directions (earlier)	10	264, +61	102	...	5
2 Abitibi dikes directions (herein)	12	267, +63	117	7	4
3 Abitibi dikes poles (herein)	12	30N, 135W	58	10	6
4 Matachewan dikes directions (earlier)					
5 Matachewan dikes directions	10	207, −16	54	10	7
6 Matachewan dikes poles	10	44N, 60E	97	8	5
11 Pike's Hill laval corrected direction	1	178, −12	(24)	(16)	(6)
12 Pike's Hill lava corrected pole	1	47N, 103E	...	...	(5)
7 Kamiskotia complex directions	7	289, +23	144	6	5
8 Kamiskotia complex poles	7	21S, 2E	140	6	5
9 Dundonald sill direction	5	269, +32	19	17	18
10 Dundonald sill poles	5	13S, 21E	17	18	19

NOTE.—In entries 1 and 4 earlier studies are summarised. In entry 1 the mean of results from 10 sites by Larochelle (1966) is given. In entries 2 and 3 the results from the two Abitibi dikes described herein are combined with those of entry 1. There are four previous results from Matachewan dikes normal and reversed directions given by Fahrig et al. (1965) and Strangway (1964). In entry 4 these four results are averaged irrespective of sign giving each unit weight $N = 4$. In this earlier work no breakdown by sites has been given so no combination with the new data given herein can be given. In entries 5 and 6 the present results are summarized from Matachewan dikes. In entries 11 and 12 the directions are corrected for bedding tilt. In entries 1, 2, 3, 5, 6, 7, 8, 9, and 10 unit weight is given to sites.

complex. Obviously this is not a satisfactory calibration of track 6 but seems the most reasonable on the basis of the present scanty evidence.

One of the main problems in Precambrian work is to determine whether the remanence directions should be referred (a) to the horizontal at the time the rock was formed on the assumption that the magnetization was acquired prior to deformation, or (b) to present horizontal on the assumption that it was acquired after deformation, or (c) to some other plane on the assumption that the magnetization was acquired at some intermediate stage of deformation. (This problem does not of course arise in studies of vertically inclined diabase dikes.) If (a) or (c) are the proper procedures, then study of deformed rocks of this very great age (and most are deformed) is severely limited, because in most situations the correct reference plane cannot be accurately determined. One of the redeeming features of Precambrian work in the Canadian Shield, which apparently makes this aspect of paleomagnetic work simpler than it might at first sight appear, is that coherent polar tracks can be obtained when directions are referred to present horizontal. This amounts to a negative fold test on a continent-wide scale. It is a feature, not only of highly deformed and metamorphosed terrains where it is to be expected (Ueno and Irving 1976; McWilliams and Dunlop 1976), but also of rocks that have undergone little or no alteration (McGlynn et al. 1974). In the present context an item of interest is the agreement between pole for the Stillwater complex (SU) and those for the Kamiskotia and Dundonald bodies, when all the directions are referred to present horizontal (fig. 16). It is therefore possible that the present horizontal (hypothesis (b)) is the correct plane of reference, and that the bodies have not undergone very much relative movement since they were magnetized. This point if substantiated by further observations, is of considerable interest, because it implies little or no relative movement between the Stillwater block and the Superior Province.

There is no polar bend (or hairpin) associated with the Kenoran Orogeny. This is different from Proterozoic and Phanerozoic orogenies that are characterized by polar loops and hairpins. This result could mean that the Kenoran Orogeny is entirely unlike Phanerozoic and

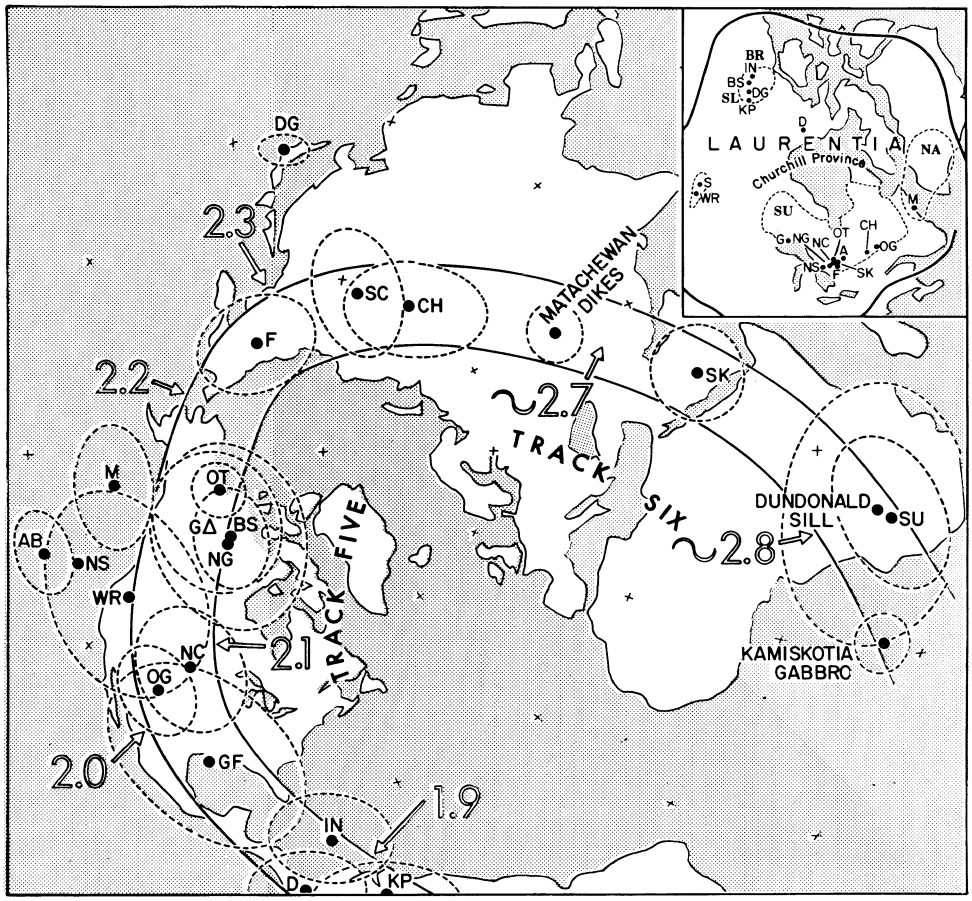


FIG. 16.—APW path for Laurentia for the interval −1.9 to −2.8 Ga. The path is intended only as a first-order interpretation of the data. The names of the new poles given in this paper are spelled out and their particulars may be found in Table 2. Other poles are assigned letters and are given below. In this list the number after the name of the rock unit is the assigned age in billions of years. The Ottawa list number refers to the Ottawa catalogues of paleomagnetic directions and poles (Hicken et al. 1972, and Irving and Hastie 1975) where details and references to the bibliography can be found. In the case of new results, not yet entered in that catalogue, reference to the original is made. AB = Abitibi dikes (Larochelle 1966 and Table 3 herein); BS = Big Spruce Complex (Irving and McGlynn 1976); CH = Chibougamou sills (Ueno and Irving 1976); D = Dubawnt Group (Park et al. 1973); DG = Dogrib dikes (McGlynn and Irving 1975); F = Firstbrook Formation Cobalt Group (Roy and Lapointe 1975); G = Gowganda Formation Cobalt Group (Symons 1975); this pole has been obtained from magnetizations acquired after folding of the Gowganda sedimentary rocks and is therefore secondary (Roy and Lapointe 1976), and it is denoted by an open triangle; GF = Gunflint (Symons 1966); IN = Indin dikes (McGlynn and Irving 1975); KP = Kahocella Group primary magnetizations (McMurray et al. 1973, Reid 1972); M = Mugford basalt (Murthy and Deutsch 1971); NS = Nipissing diabase (Symons 1970, 1971); NC = Nipissing diabase (Roy and Lapointe 1976); NG = Nipissing diabase (Symons 1975); OG = Otish gabbro (Fahrig and Chown 1973); OT = Otto stock (Pullaiah and Irving 1975); SC = Stillwater complex (corrected) with respect to layering (Berg 1970); SK = Skead Group (Ridler and Foster 1975); SU = Stillwater Complex (uncorrected with respect to present horizontal, data of Berg (1970) recalculated in Irving and Hastie (1975); WR = Wind River dikes (Spall 1971).

Proterozoic orogenies, or it could simply reflect the inadequacy of the record available at present. There does, however, appear to be a hairpin at about −2.3 Ga and this could be the polar signature of a mid-Aphebian orogeny. Further evidence for this is to be found in the divergence of the Dogrib pole (−2.6 Ga) from track 6. If this is indicative of movement between the Slave and Superior Provinces then this movement appears to have been mid-Aphebian in age and was completed by about −2.1 Ga, because poles of this age from Slave and Superior available at the present time and are in substantial agreement (McGlynn et al. 1975).

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